

Virial theorem: $V^2 = \alpha \frac{M}{R}$

$$\frac{M_{gal}}{L} = \alpha \frac{R \sigma_r^2}{G L}$$

$$\mu \Rightarrow L = \frac{\alpha R \sigma_r^2}{\mu G}$$

$$L \propto R^2 I \Rightarrow R \propto \sqrt{\frac{L}{I}}$$

$$L = \frac{\beta L^{1/2} \sigma_r^2}{\sqrt{I_0} \mu G}$$

$$L = \frac{\gamma \sigma_r^4}{\langle I_0 \rangle \mu^2} = \tilde{\gamma} \frac{\sigma_r^4}{\langle I_0 \rangle}$$

$$R_e^2 = \frac{\omega \sigma_r^4}{I^2}$$

$$2 \ln(R_e) = \ln(\omega) + 4 \ln(\sigma_r) - 2 \ln(I)$$

$$\ln(R) = C + 2 \ln(\sigma_r) - \ln(I_e)$$

$$\ast \mu = \alpha L^\beta$$

$$\alpha L^{\beta+1} = R \sigma_r^2$$

$$\tilde{\alpha} (R^2 I)^{\beta+1} = R \sigma_r^2$$

$$\ln(\tilde{\alpha}) + \beta+1 \ln(R^2) = -(\beta+1)$$

$$\tilde{\alpha} \left(\frac{2\beta+1}{D} \right) = \frac{\sigma_r^2}{I^{\beta+1}}$$

$$\ln(R) = \frac{2 \ln(\sigma_r)}{2\beta+1} - \frac{(\beta+1) \ln(I)}{2\beta+1}$$

$$\underline{1.24} = \frac{2}{2\beta+1} \Rightarrow 2\beta+1 = \frac{2}{1.24} \quad \downarrow .81$$

$$\beta \approx 0.30645$$

$$\beta \approx .8$$

$$\frac{V^2 R \alpha}{L} = \frac{\mu}{L} \Rightarrow \frac{V^2 R \alpha}{\mu} = L$$

$$\text{then use } \mu \sim L^\beta$$

$$\ast \frac{V^2 R \tilde{\alpha}}{L^\beta} = L \Rightarrow L^{\beta+1} = V^2 R \tilde{\alpha}$$

$$(R^2 I_e)^{\beta+1} = V^2 R \hat{\alpha}$$

$$R^{2\beta+1} I_e^{\beta+1} = V^2 \hat{\alpha}$$

$$(\beta+1) \ln(R) = (2/2\beta+1) \ln(V) - \frac{\beta+1}{2\beta+1} \ln(I)$$

$$+ \alpha$$

